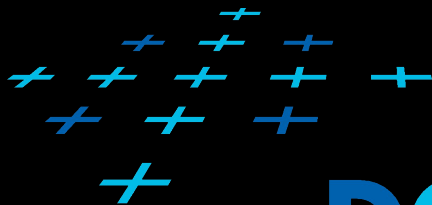


VIZ 13

VISUAL ANALYTICS AND BIG DATA

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Faculty of Electrical Engineering at CTU in Prague

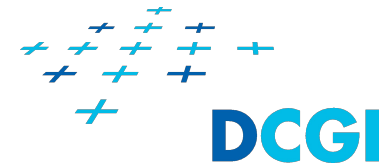


DCGI



- + Introduction
 - + What is visual analytics
- + Big data
 - + What are big data
 - + V3(4)
 - + Volume, Variety, Velocity, (Veracity)
 - + Challenges in big data visualization
 - + Some solutions
- + Automated analysis techniques + visualization
- + Visual analytics examples

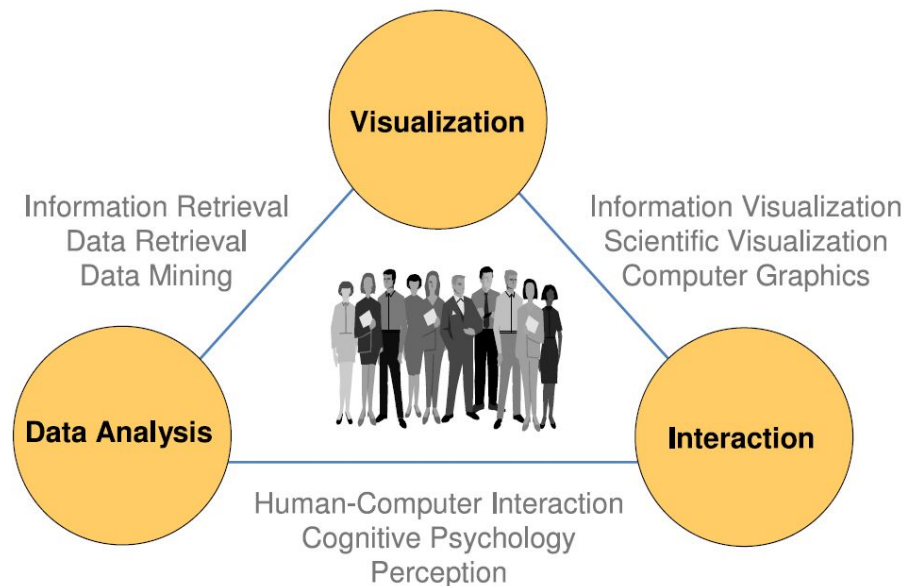
Introduction



- + Visual analytics combines **automated analysis techniques** with **interactive visualizations** for an effective understanding, reasoning and decision making on the basis of **very large and complex datasets (big data)**
- + Visual analytics combines three main ingredients
 - + Visualization
 - + Data mining
 - + Interaction
- + You may encounter different term with similar meaning to visual analytics
 - + Visual data mining - visualization and data mining, less or no interaction
 - + etc.

- + Visual analytics is the interplay between
 - + Visualization
 - + High-bandwidth input information channel to brain used effectively
 - + Data mining
 - + Advanced computational capabilities
 - + Interaction
 - + Allows to control the visualization and data mining

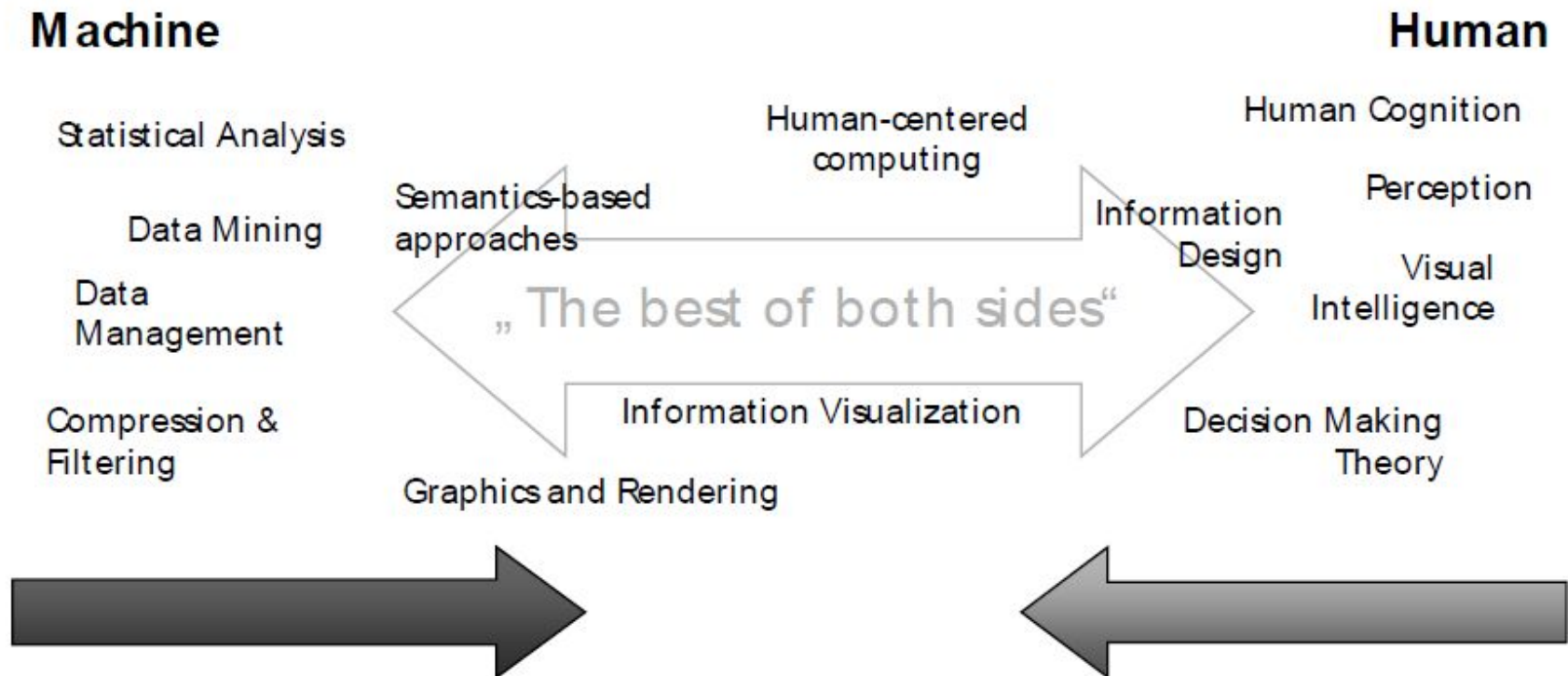
to augment the analysis capabilities of humans



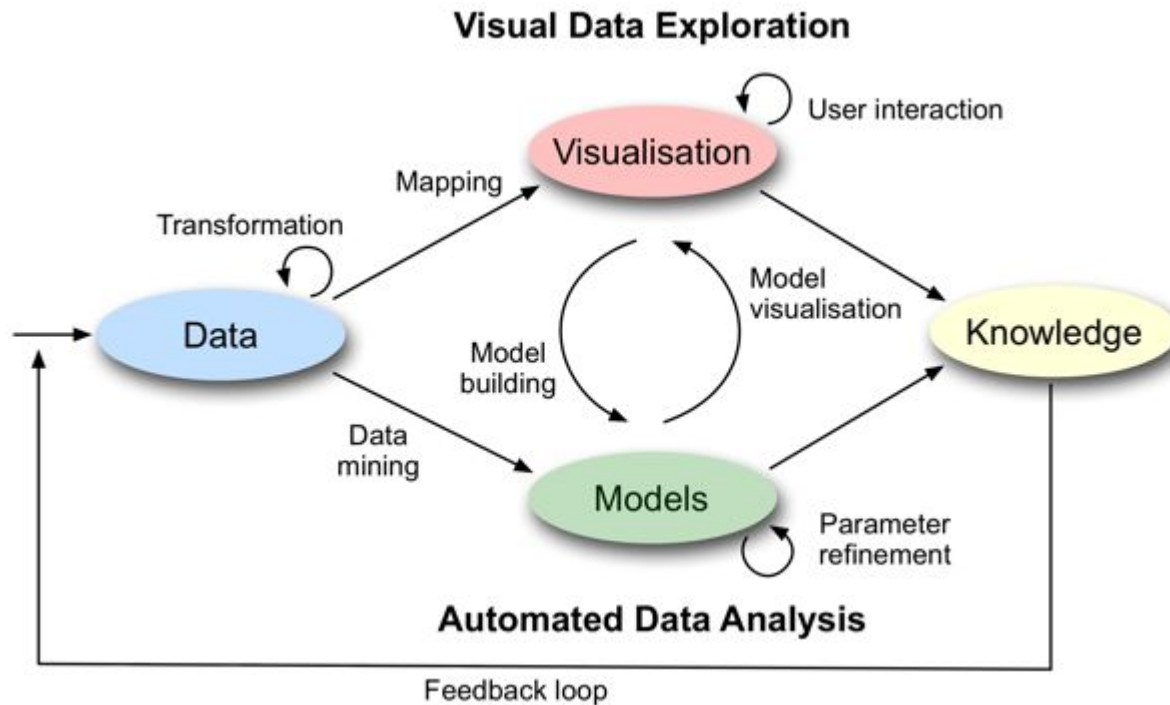
- + Are enabling the **analyst** to use their full perceptual and cognitive abilities on the analytical processes
- + Are allowing the **analyst** to augment their discovery process by applying advanced computational capabilities
- + Who is the analyst?
 - + In the case of visual analytics, the analyst is typically **not an expert** in data mining and/or visualization
 - + It is a **decision maker** that is making decisions based on the **analysis of data** in **limited time**
 - + Head of fire brigade, disaster manager, etc.
 - + Member of the team of decision maker
 - + Member of town hall, etc.
 - + Need to **disseminate/present** results of the analysis to the decision maker

Visual analytics

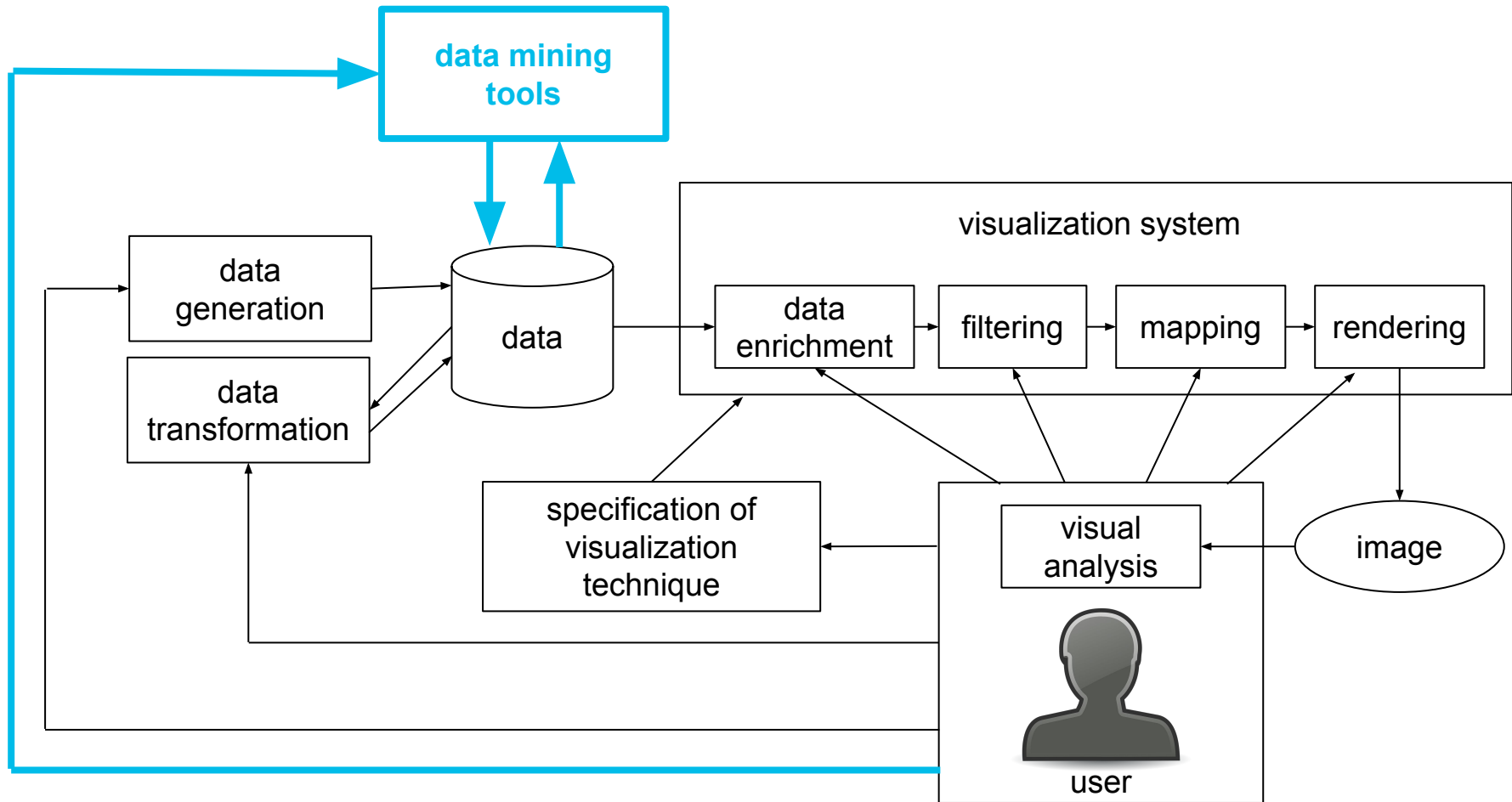
- + Visual analytics is dividing the labour between human and machine to get the best from the both



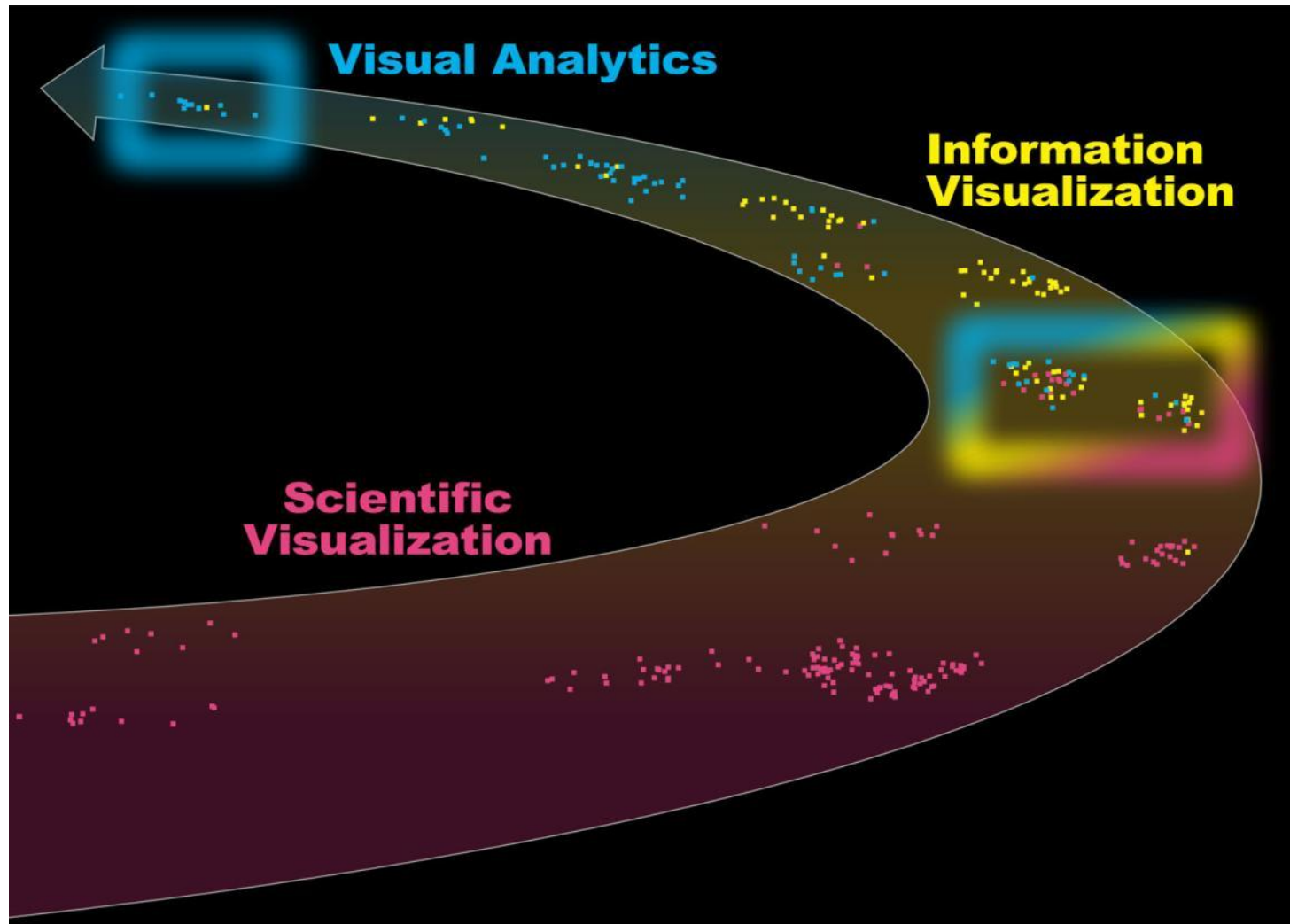
Visual analytics process



Reference model for visual analytics



The Landscape of Visualization Science



Publications from IEEE VisWeek, 2006, 2007, 2008

In the following slides we will

1. Define what are big data and
2. Look at techniques that might help us visualize big data
 - + Most of them use **automated analysis techniques** in combination with **visualization**
3. Show two examples of visual analytics applications
 - + First for abstract data
 - + Second for spatio-temporal data

BIG DATA

+ What is Big Data?

- + Well, lots of data, right?
- + A new buzz-word! ... but a relevant one!

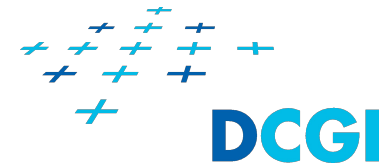
+ Examples

- + Big data from large sensor networks (Internet of Things, ...)
- + Big data from healthcare (National Health Database, ...)

+ Broadly used definition

- + **3V-definition:** Big data is **high-volume**, **-velocity** and **-variety** information assets that demand cost-effective, innovative forms of information processing for enhanced insight and decision making.
[Doug Laney, 2001 / Gartner]

Big Data, V#1: Volume

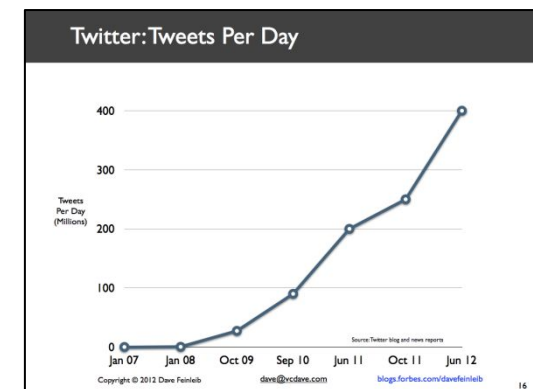
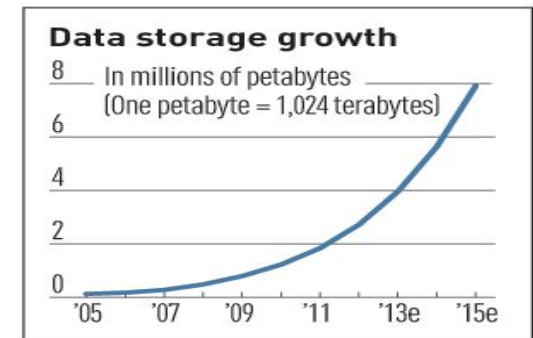


- + *Big data* refers to datasets whose size is beyond the ability of typical database software tools to capture, store, manage, and analyze.

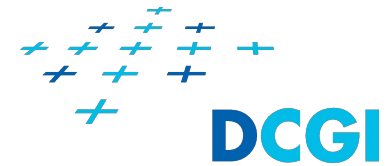
[McKinsey Global Institute 2011]

- + Available data grows exponentially

- + Exabytes of data available world-wide
 - + 1EB = 1000 PB = 1 million TB = 1 billion GB
 - + hundreds of EB transferred via the Internet, annually
 - + EB of new information stored, annually



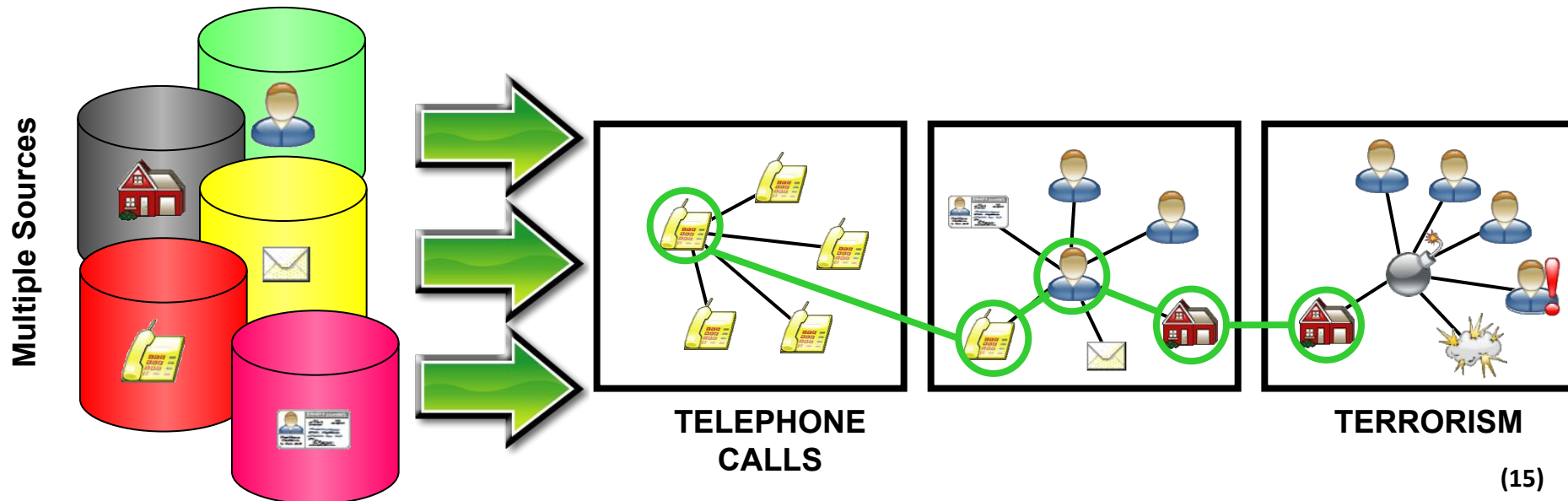
Big Data, V#1: Volume



- + 30 billion pieces of information on Facebook per month!
400 million tweets per day
4 billion hours of videos are watched on YouTube / month
>400 million wearable, wireless health monitors
- + Daniel Keim, 2007:
100 million FedEx transactions per day,
150 million VISA credit card transactions per day,
300 million long distance calls in ATT's network per day,
50 billion e-mails worldwide per day,
600 billion IP packets per day DE-CIX backbone
- + Dark Data: available, but unused data

Big Data, V#2: Variety

- + Data beyond numbers
 - + text, images, and sound, relational data, ...
unstructured data
- + Data from multiple sources
 - + Data from different sources can be incompatible or contradicting
 - + Data management (also called data wrangling) is needed
 - + The process of data transformation from raw data to data usable for analysis - done by human using specialized applications
 - + This is a huge problem with big data



Entity Resolution

Source A



Jonathan Q. Adams
Boston, MA
05/29/1968
DL:54321-123

Source B



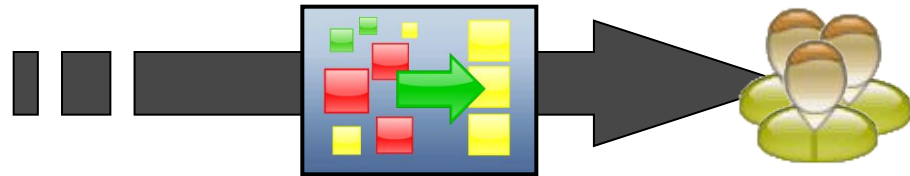
John Quincy Adams
123 Main Street
Boston, MA
774-207-0000

Source C



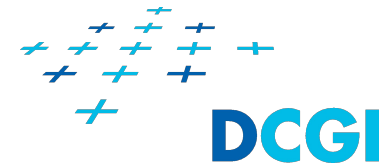
Quincy Adams
Bedford, MA
774-207-0000
12/05/1965

Are they the
same entity?



*Tokenizing
Standardization
Normalizing
Aliasing
Permutation*

Big Data, V#3: Velocity



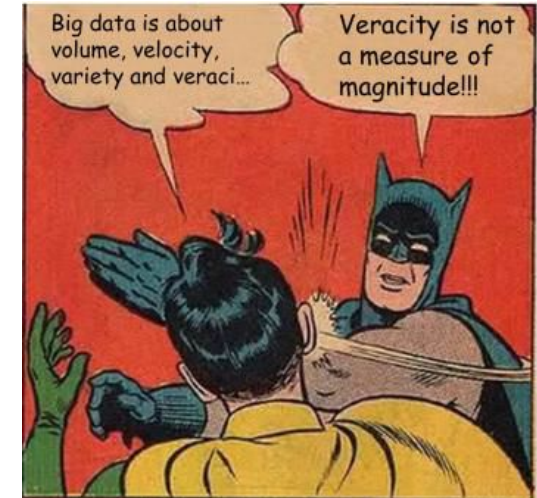
- + Real-time Big Data / Streaming Data Analysis
 - + rapidly changing data
 - + data at different speeds and uneven rates (bursts)

- + Big Data – a moving target!
 - + lots of generated information cannot be stored!
 - + 90% of healthcare data is discarded (videos, etc.)

Big Data, V#4?: Veracity

- + Uncertain / low-quality data
 - + >\$3 trillion loss to US economy due to bad data quality
 - + high degree of uncertainty

- + Even more Vs: [K. Normandeau]
 - + Validity: the right data for the right decisions?
 - + Volatility: when valid, storing for how long, etc.?



Challenges in Big Data Visualization

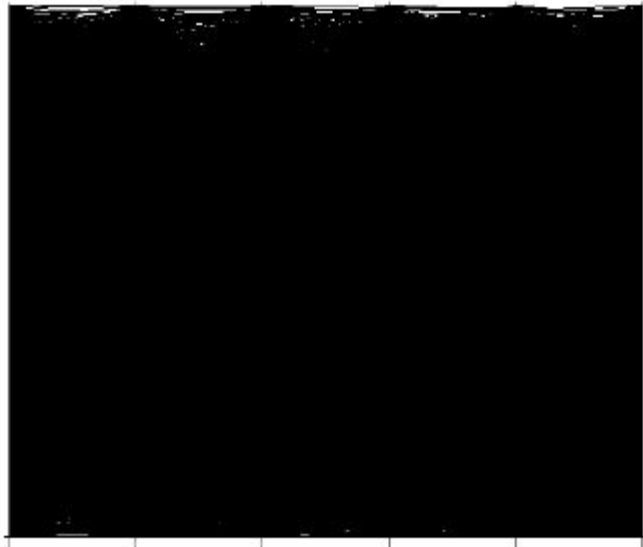


- + Big data visualization $\neq$ straight-forward
- + Significant practical challenges
 - + Technical limitations (memory, bandwidth, ...)
 - + Distributed out-of-core solutions (cloud, ...)
 - + Managing big data architectures (Hadoop, ...)
- + Visualization challenges
 - + **Conceptual challenges (invalid principles, ...)**
 - + User-side challenges (perceptual, cognitive, ...)
 - + Our perceptual and cognitive abilities are not improving

Conceptual challenges

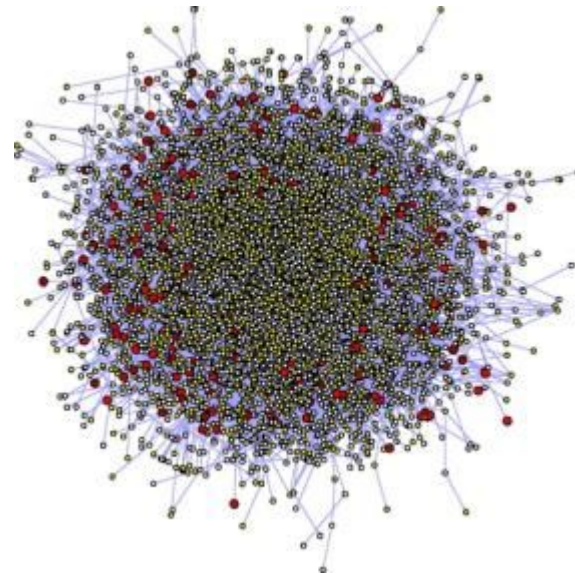
- + Scaling problem (in addition to technical scaling problems)
 - + Techniques that work for medium-sized data break down for large or huge data
- + New concepts for visualization of big data are needed

Parallel coordinates



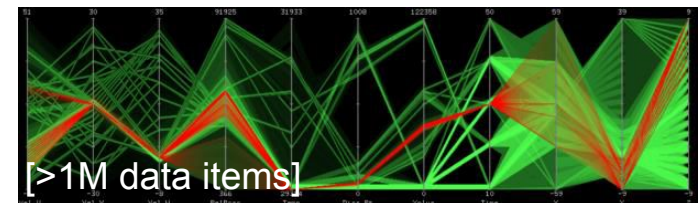
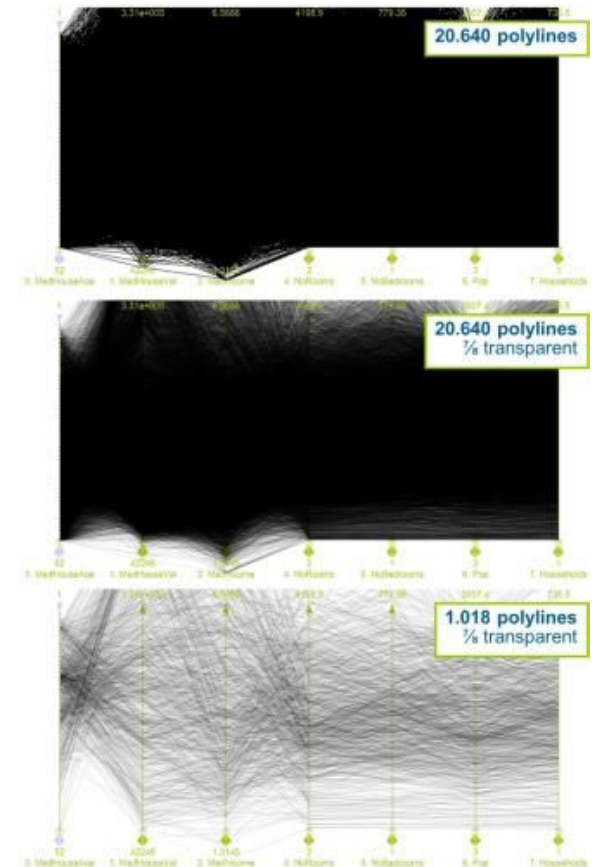
15,000 data items with noise

Graph visualization



Meeting Big Data Visualization challenges

- + Technological optimization
 - + Optimized software
 - + Parallelization, etc.
- + Adapting existing visualization techniques
 - + Semi-transparency against overplotting
 - + Edge bundling for large graphs
- + Conceptual change
 - + Frequency-based visualization
 - Binning*
 - + Hierarchical exploration
 - Clustering*
 - Multi-scale representations*
 - + Hybrid approaches
 - Automated data analysis + visualization*



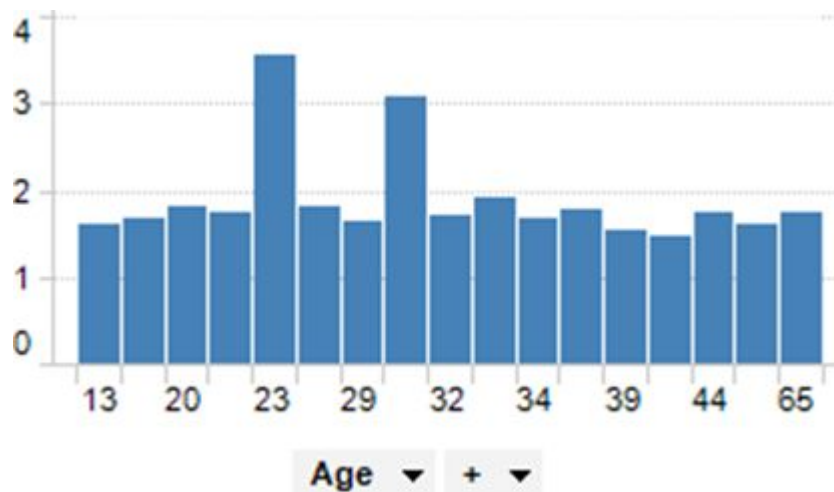
FREQUENCY-BASED VISUALIZATION

Binning

Frequency-based visualization

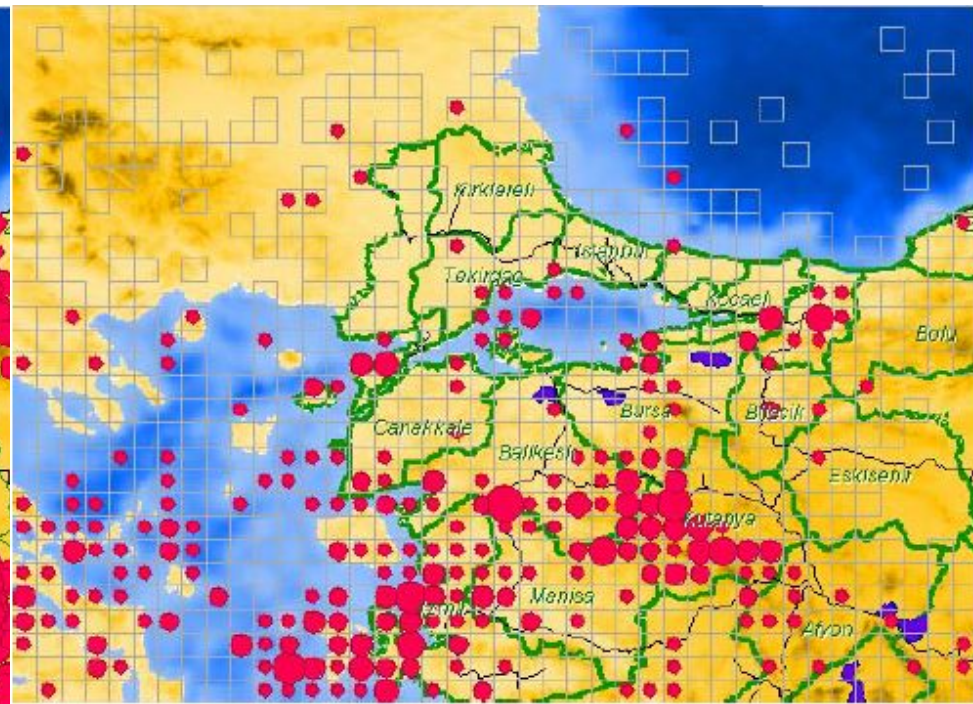
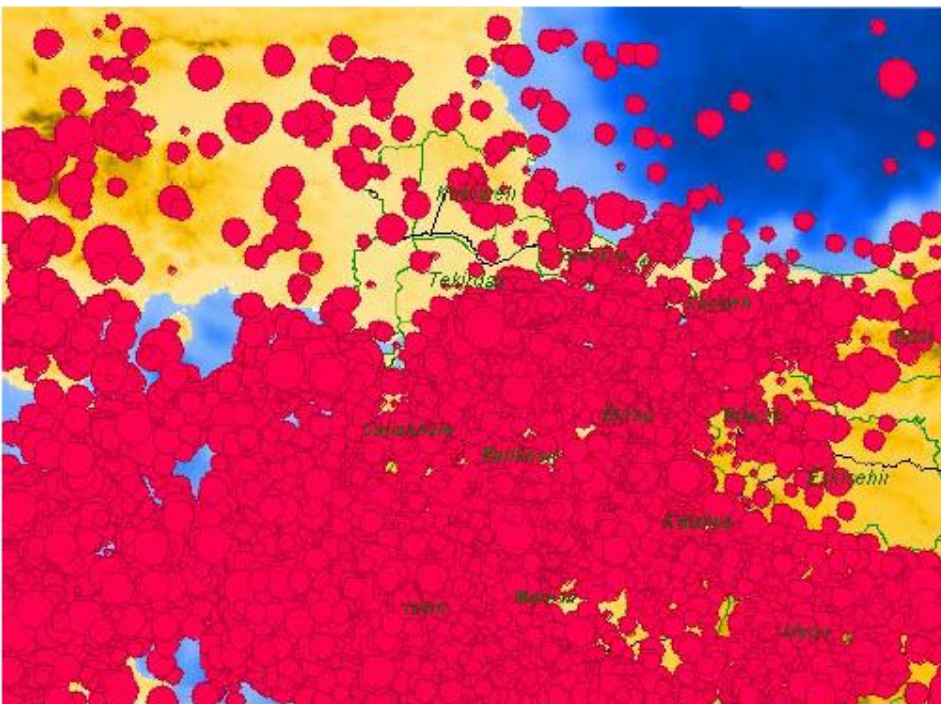
+ Binning

- + A way to group values of quantitative attribute into a number of "bins"
- + Typically, we count number of items (frequency) in each bin
- + Other functions are possible - maximum value, average value, etc.



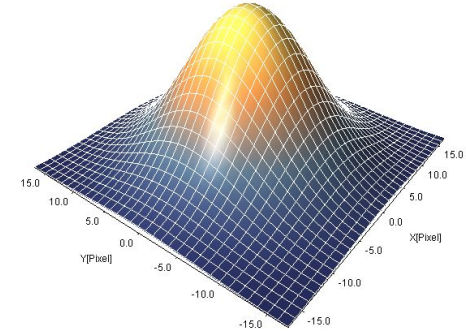
Examples of frequency-based techniques

- + Earthquakes that occurred in the area between Greece and Turkey
- + When we visualize all data instances, we don't see any patterns
- + When we use binning and show frequency, the patterns emerge



Examples of frequency-based techniques

- + Density based visualization of traffic accidents
 - + We represent each accident as a gaussian kernel
 - + The gaussian kernels overlap, for each pixel we count sum of the overlapping kernels
 - + We map the sum on color



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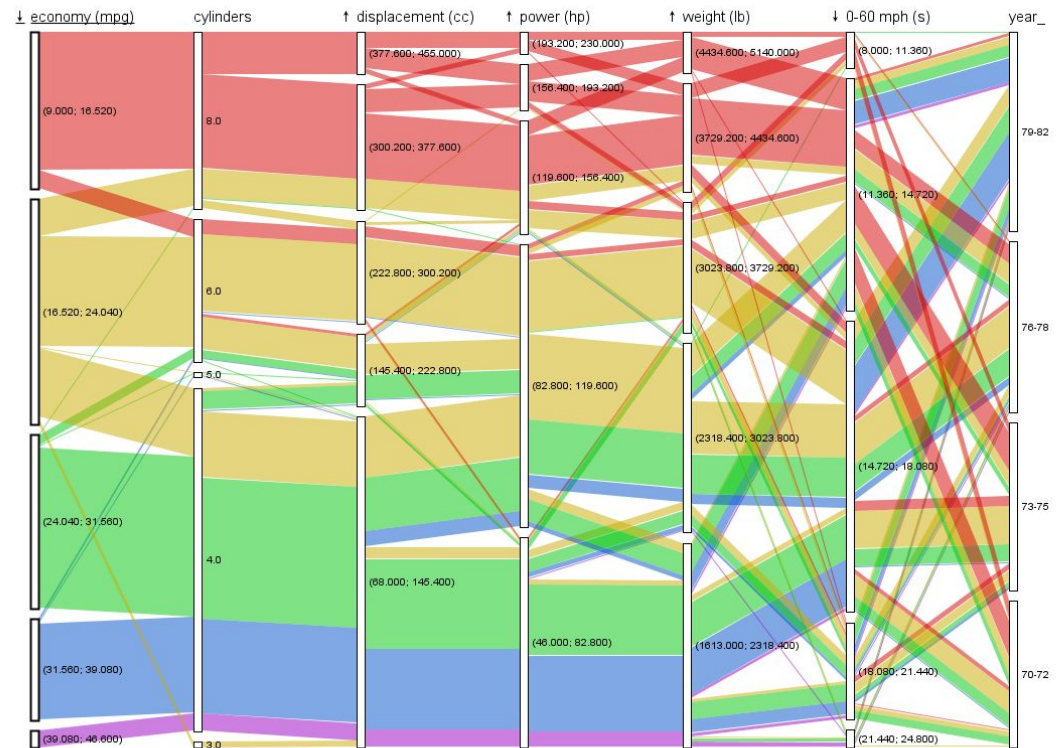
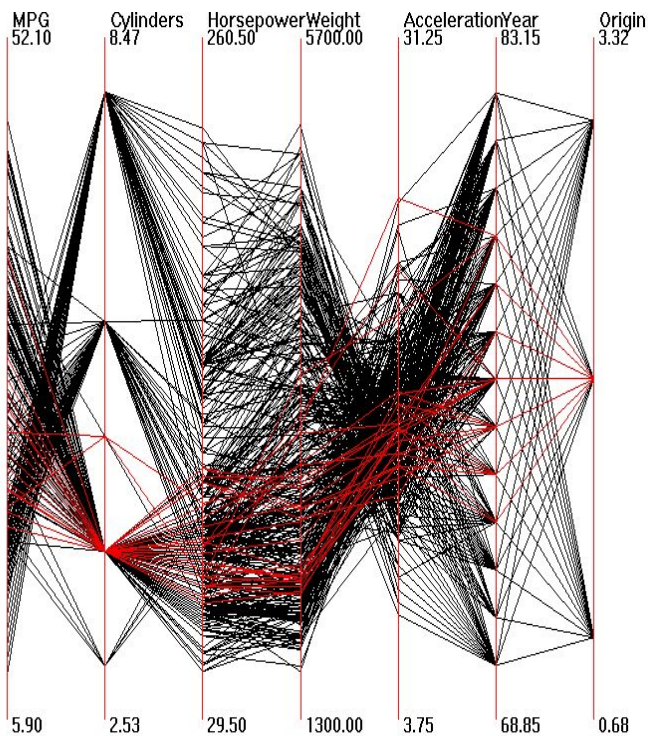
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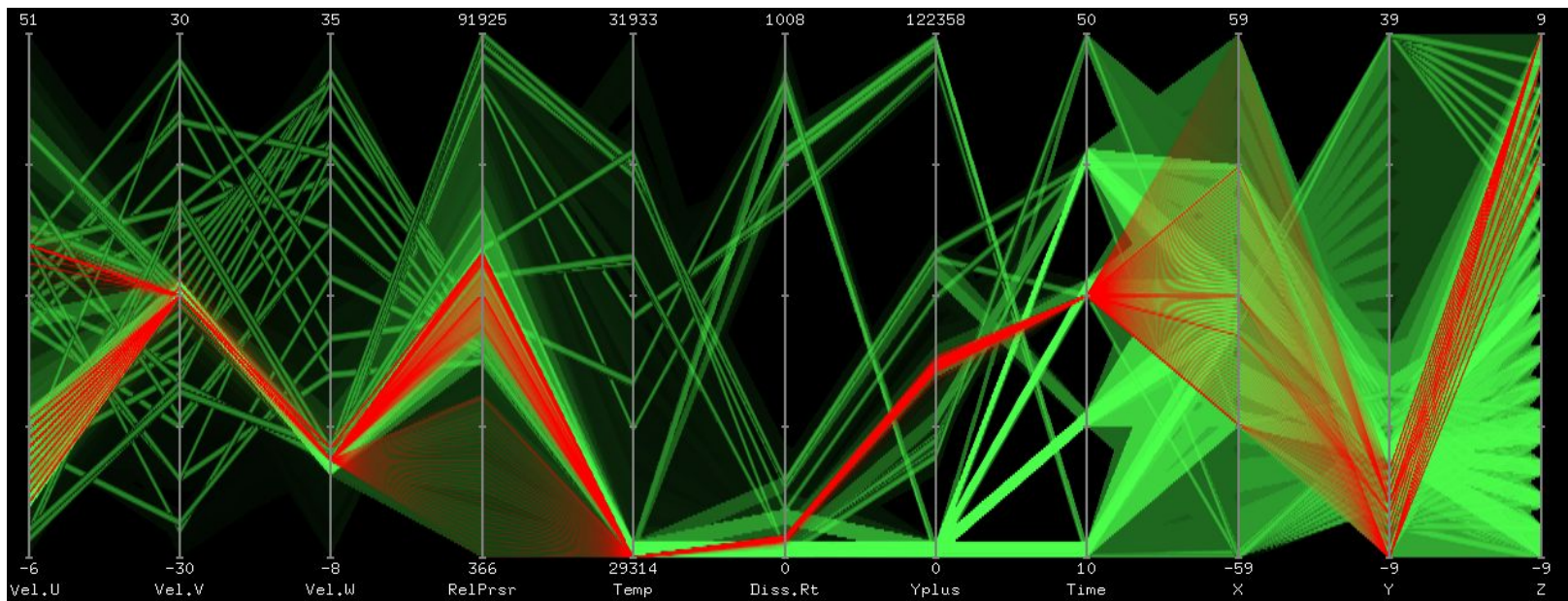
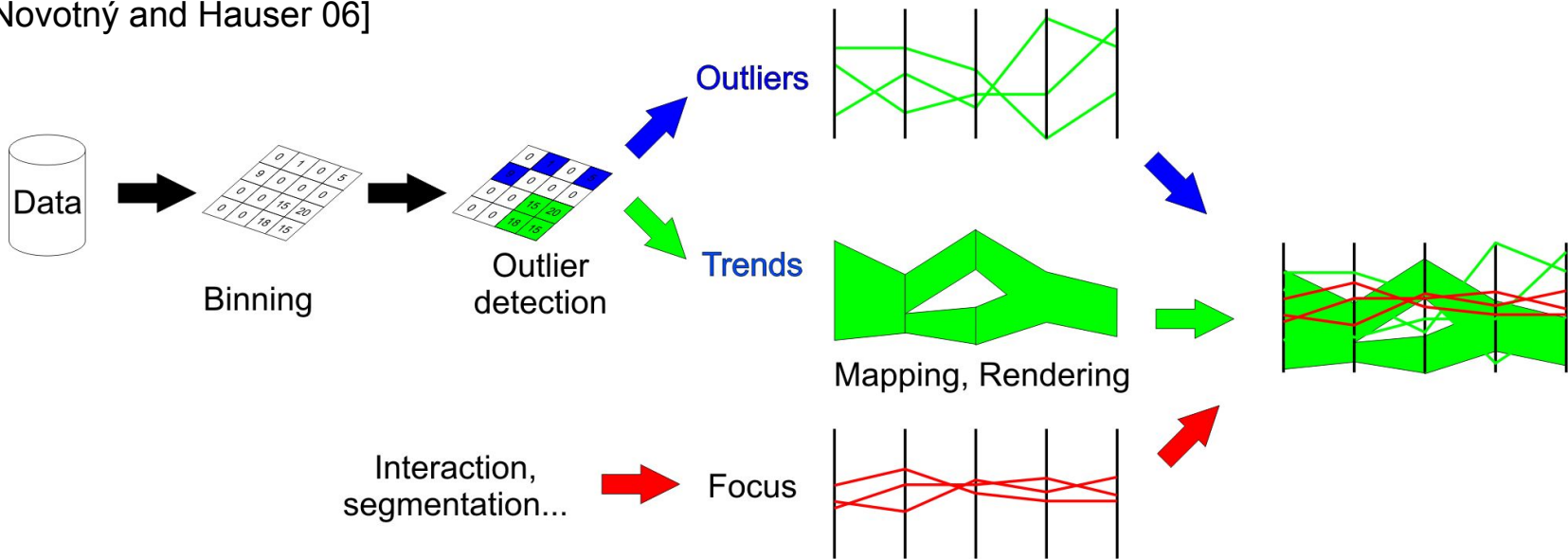
Examples of frequency-based techniques

Parallel sets



Outlier-preserving focus+context visualization in parallel coordinates

[Novotný and Hauser 06]



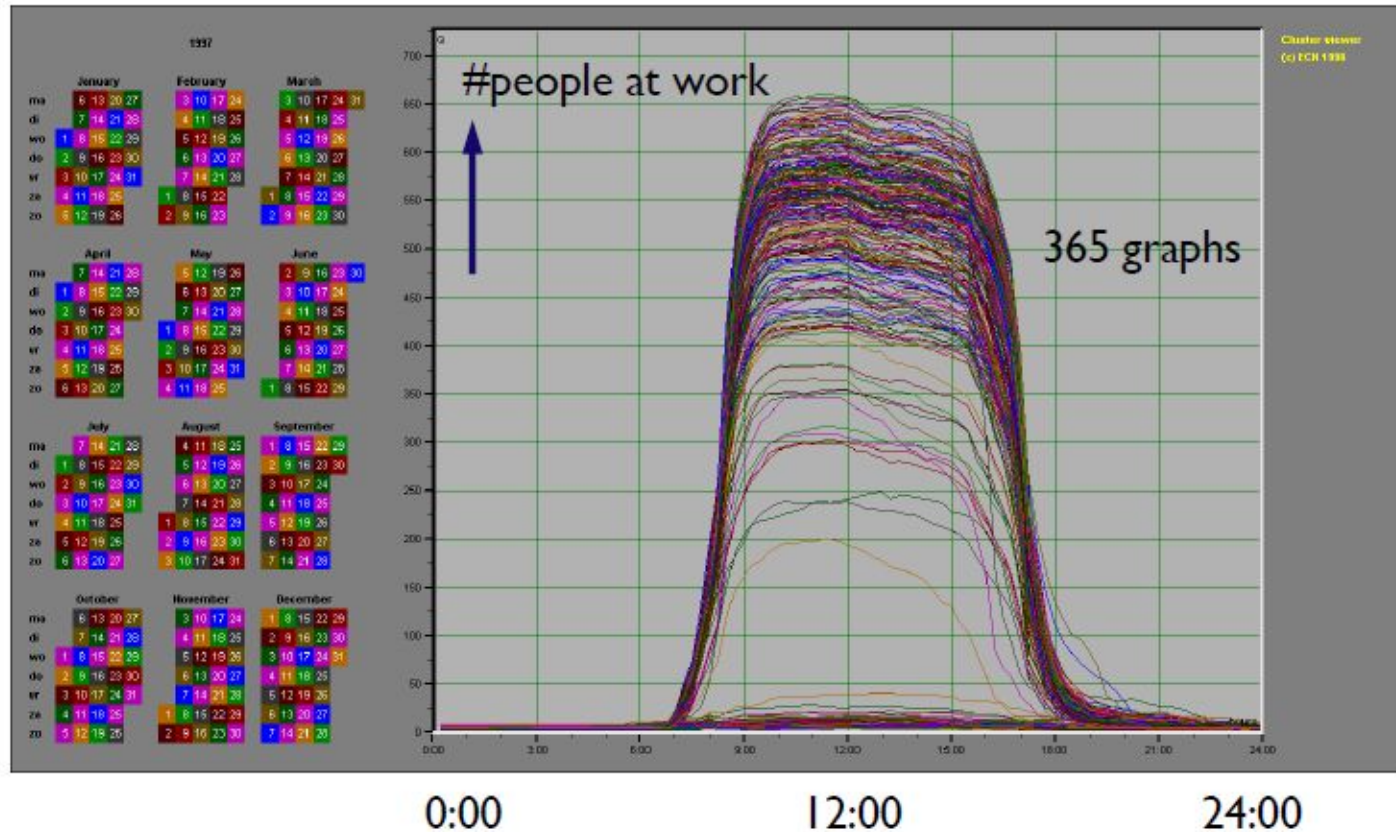
HIERARCHICAL EXPLORATION

Clustering

- + **Clustering** is the task of grouping a set of objects in such a way that objects in the same group (**cluster**) are more similar (according to certain metric) to each other than to those in other groups (clusters).
- + There are many clustering methods
 - + K-means clustering
 - + Mean-shift clustering
 - + Density-Based Spatial Clustering
 - + Expectation–Maximization Clustering
 - + Agglomerative Hierarchical Clustering
 - + etc.
 - + See the link for description of the algorithms and their comparison
<https://towardsdatascience.com/the-5-clustering-algorithms-data-scientists-need-to-know-a36d136ef68>

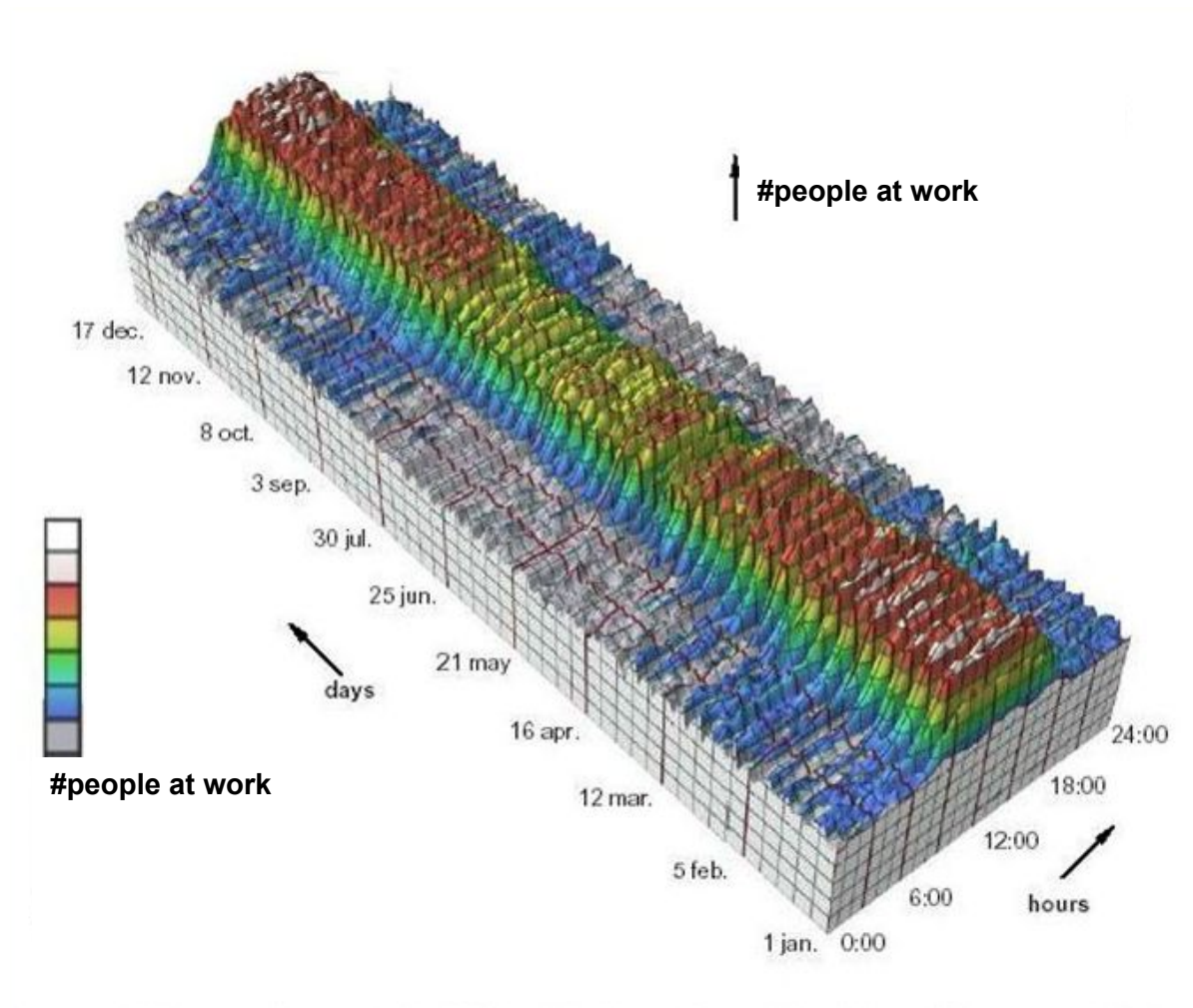
Use of clustering in visual analytics

One year of time-series data



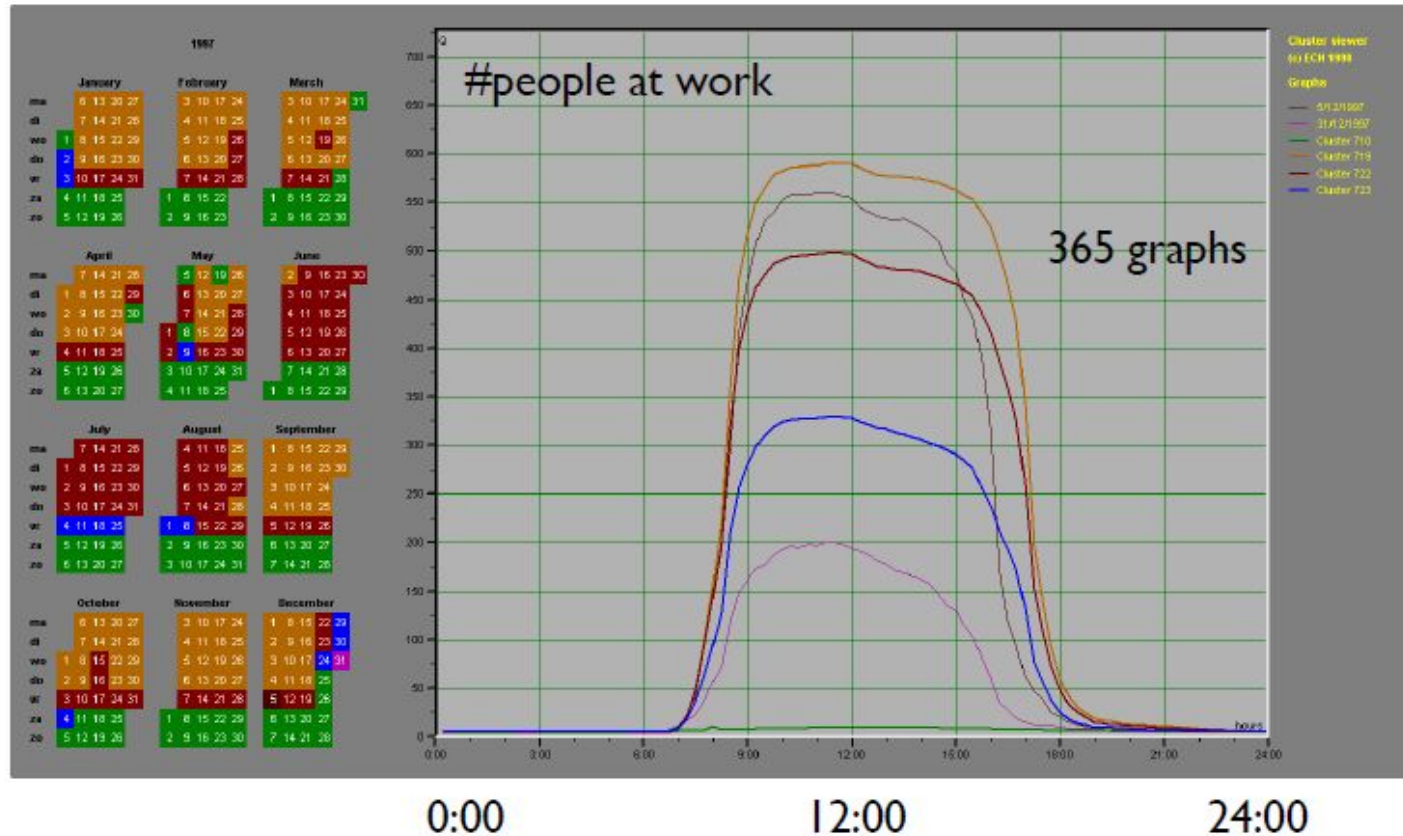
Van Wijk et al., 1999

Calendar based visualization



Use of clustering in visual analytics

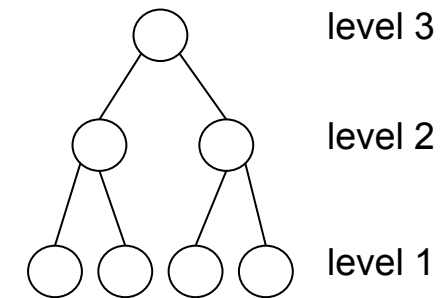
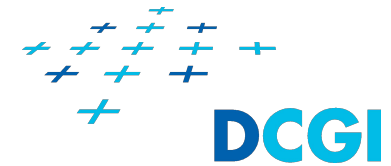
After clustering



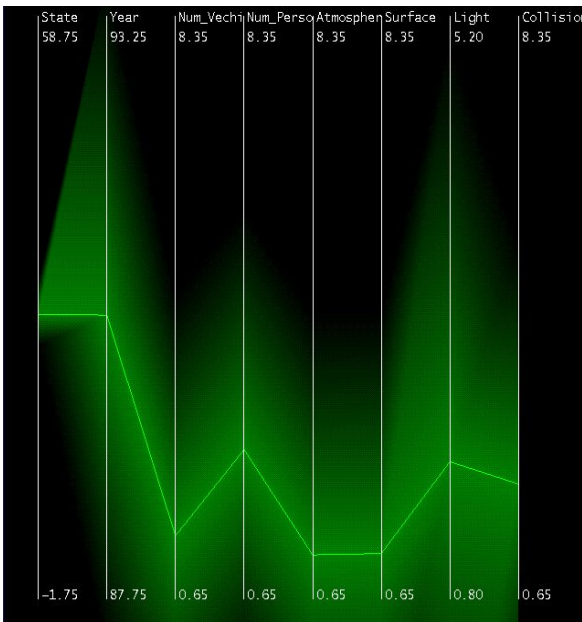
Van Wijk et al., 1999

Hierarchical exploration

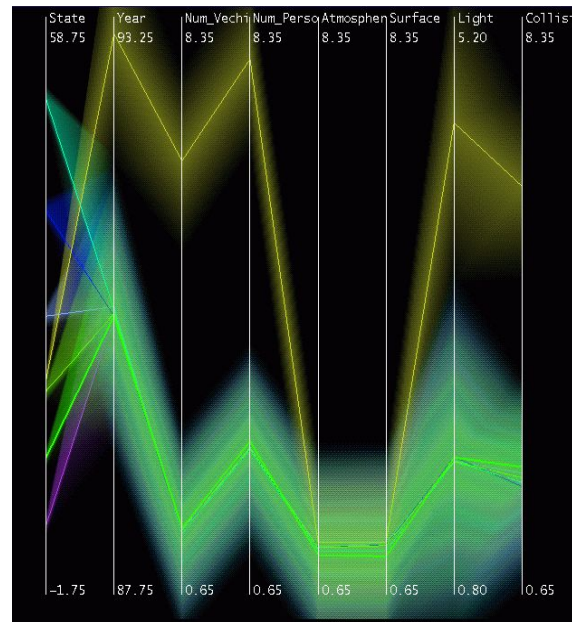
- + Hierarchical parallel coordinates for exploration of large datasets [Fua et al. 99]
- + Visualizations for various levels of the cluster hierarchy



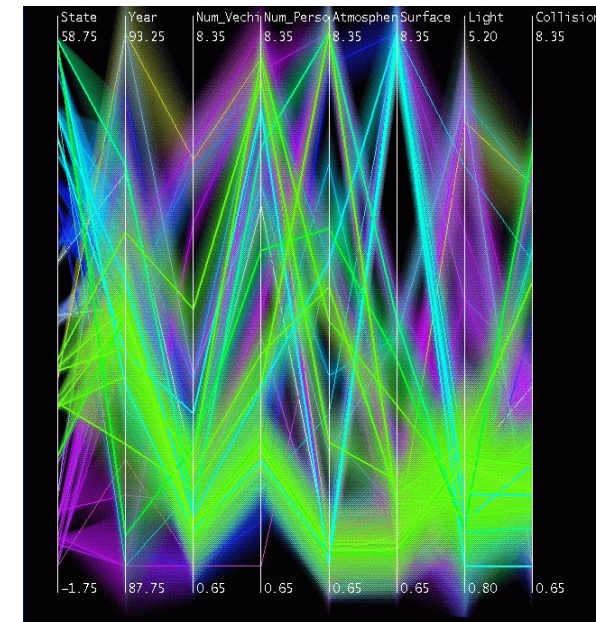
Cluster hierarchy



level 3 - root



level 2



level 1

AUTOMATED ANALYSIS TECHNIQUES + VISUALIZATION

Visual Data Mining vs Visual analytics

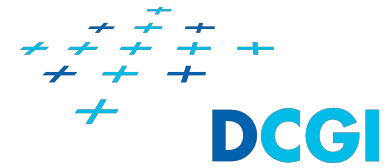
Regression

Box-plots

Confidence intervals

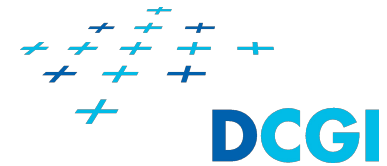
Intrusion detection

What is data mining?



- + Knowledge discovery from data
- + Extraction of interesting (non-trivial, previously unknown and potentially useful) patterns from huge amounts of data
- + Alternative names:
 - + Knowledge discovery in databases (KDD), knowledge extraction, data/pattern analysis, data archeology, data dredging, information harvesting, business intelligence, etc.

Data mining techniques



- + Regression (linear, polynomial, ...)
 - + Essentially a construction of a model of correlation/dependency of the data attributes
- + Clustering
 - + Aggregation/Reduction of the data items
 - + Construction of hierarchical structure of the data set
- + Box plots/Confidence intervals
 - + Summarization of the dataset with statistical methods
- + Outlier/Intrusion detection
 - + Detection of data items that do not fit to the model or are different from average data item
- + Decision trees and neural networks
 - + Classification of the data items into categories
 - + Represent model of the data

+ Visual Data Mining

- + New approach for exploring very large data sets
- + Combination of traditional mining methods and visualization techniques

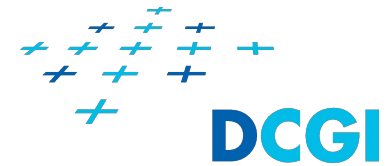
+ Advantages

- + Combination of strengths of data mining and visualization

+ Challenges

- + User needs to select the appropriate data mining technique in a given situation
- + Find visualization techniques suitable for the results of data mining operations
- + User needs to be experienced both in data mining and in visualization

Visual data mining vs visual analytics



+ Visual data mining

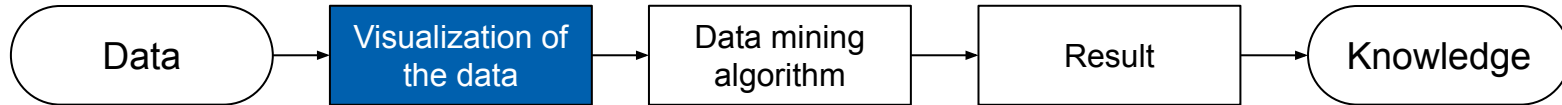
- + Computer performs data analysis, human uses the result
- + Visualization is used
 - + (mainly) to help the human understand the results
 - + (sometimes) to select appropriate data mining technique and prepare input data (data management/wrangling)
- + User is experienced in data mining and visualization
- + Interaction is typically not necessary

+ Visual analytics

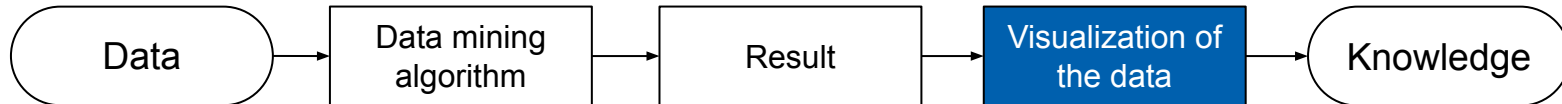
- + Human solves a complex problem with a software that helps her/him
- + They are iteratively cooperating together to find solution to the problem
- + Human is interactively exploring the data, improving the model based on new findings, then the new model may again lead to new findings, etc.
- + Interaction is required - both data mining and visualization needs to be interactive
- + Human is not an expert in data mining nor in visualization
- + The expertise is encoded in the software

Visual data mining vs visual analytics

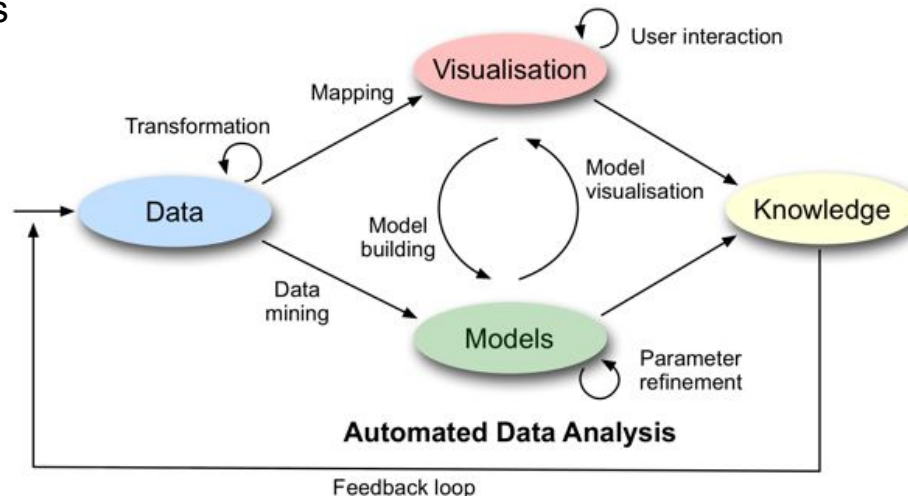
Visual data mining process - Preceding Visualization



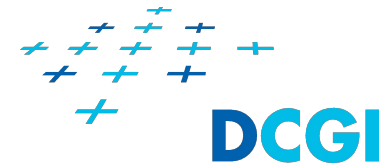
Visual data mining process - Subsequent Visualization



Visual analytics process



Data mining techniques in visual analytics



- + Regression (linear, polynomial, ...)
 - + Essentially a construction of a model of correlation/dependency of the data attributes
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 - + Aggregation/Reduction of the data items
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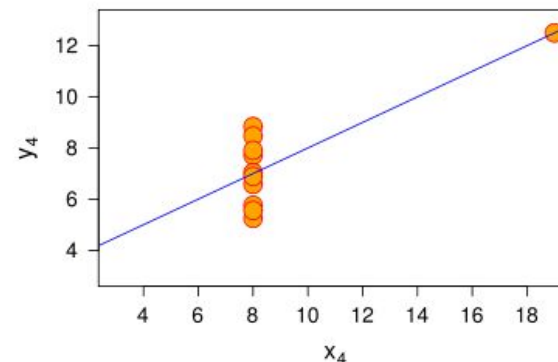
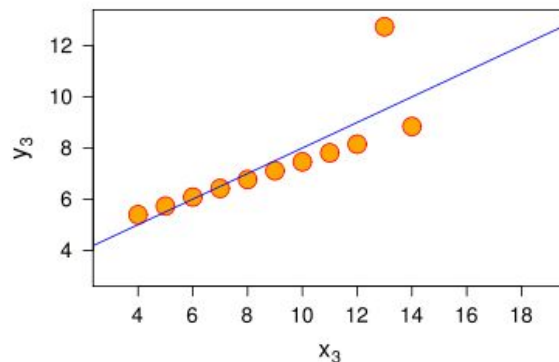
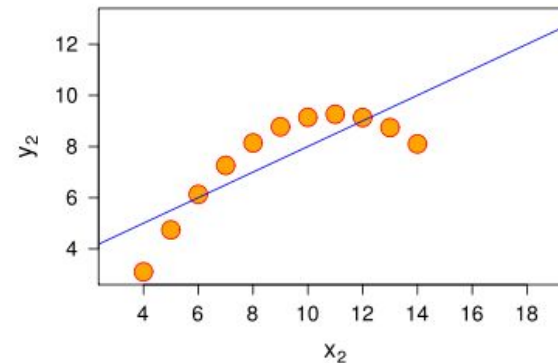
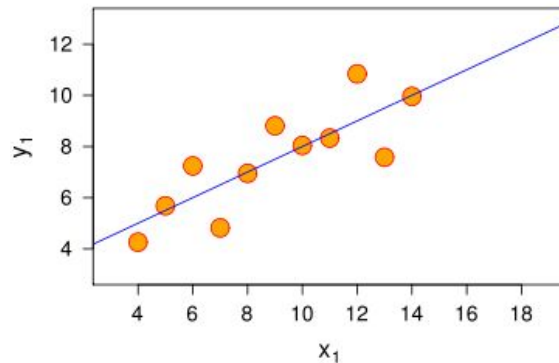
Regression

- + Model of correlation between years of experience and salary obtained with linear regression
- + By visualizing the data and the model together we see
 - + If the model fits the data
 - + If there are some outliers



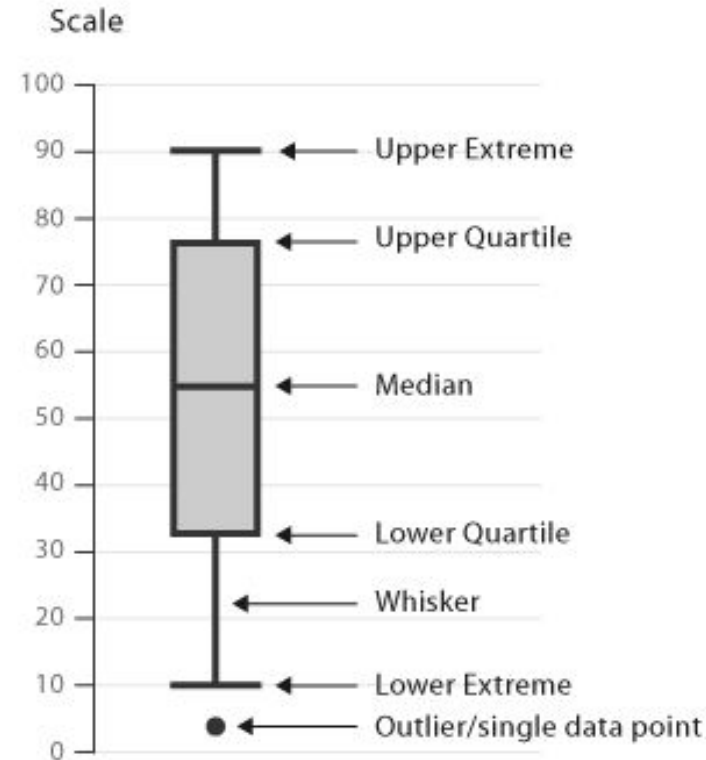
Regression

- + With visualization of the model, we can figure out much quicker if the model does not fit (e.g., correlation is not linear) or if there are outliers (that influences calculation of the model)



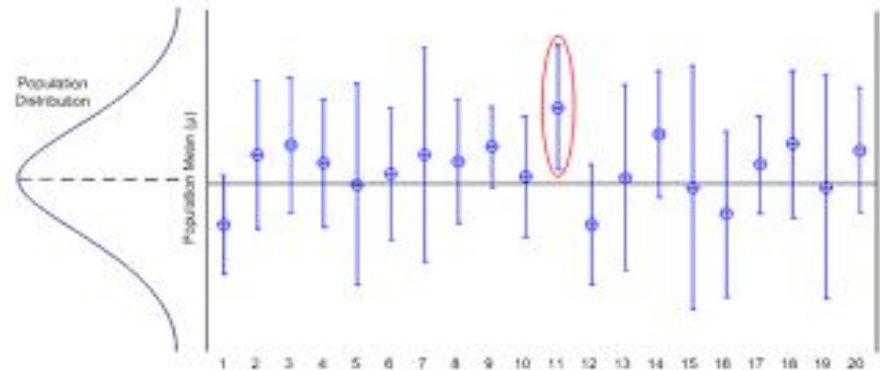
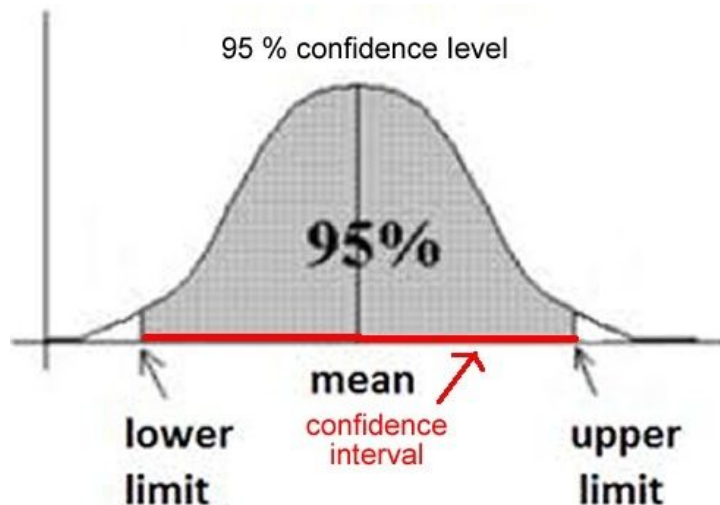
Box plots

- + Boxplots are a way of summarizing data through visualizing the five number summary which consists of
 - + the minimum value,
 - + first quartile,
 - + median,
 - + third quartile,
 - + and maximum value of a data set.
- + The boxplot divides the data items (except outliers) into 4 groups where in each is 25% of the data items
- + Boxplots visualize distribution of the data items and if the distribution is skewed or not



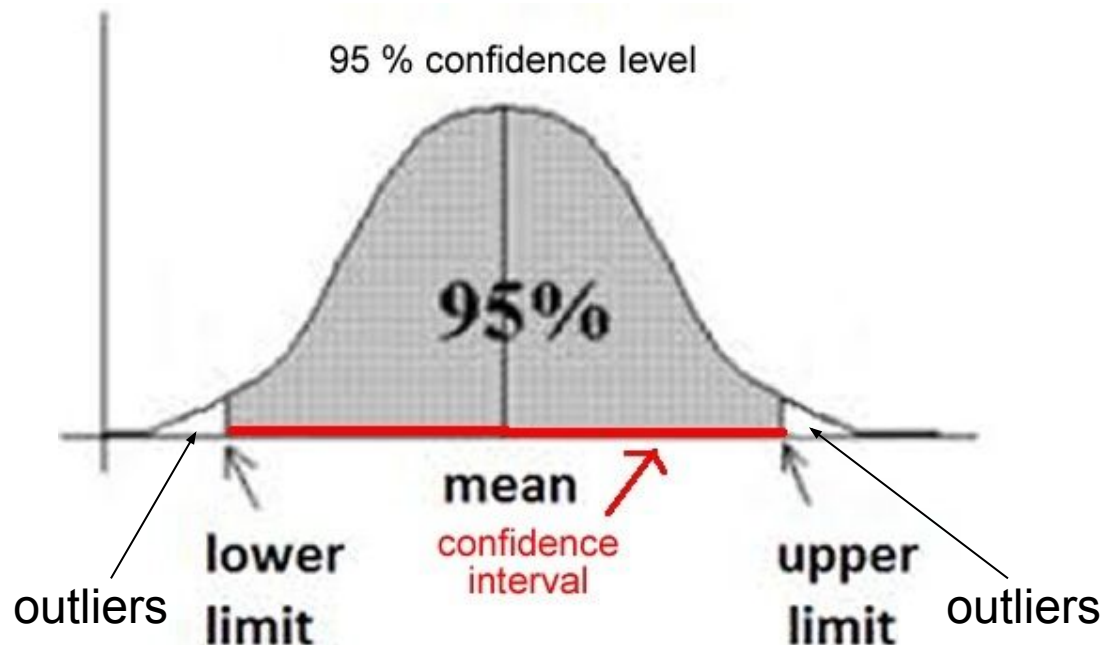
Confidence interval

- + Interval that with certain confidence contain the true mean of the population
 - + We have random samples from the population and calculate the interval from the samples
 - + In our case the samples are our data items
 - + With fewer samples we get wider confidence interval
- + Confidence level (typically 90%, 95% or 99%)
 - + If we calculate 95% confidence interval 100 times (each time we choose different samples from the population) then ~95 of the intervals will contain the true value of the population and ~5 will not



Confidence intervals

- + Are used used for null hypothesis testing
 - + We can compare means of two independent groups (between-subject design) and of two dependent groups (within-subject design)
 - + They are visual (and more intuitive) variant of p-values
- + Can be used for outlier detection



Outlier/Intrusion Detection



- + Detection of data items that do not fit to the model or are different from average data item
 - + Such data items are suspicious and should be analyzed more closely
- + Signature-based detection
 - + Building predictive models from labeled data sets (instances are labeled as “normal” or “outlier”) to identify outliers
 - + The predictive model is typically decision tree or trained neural network
 - + High accuracy in detecting many kinds of known outliers (those in labeled data sets)
 - + It may not detect unknown outliers
- + Anomaly detection
 - + Detect anomalies as deviations from “normal” behavior
 - + Potential high false alarm rate - previously unseen (yet legitimate) system behaviors may also be recognized as anomalies

Signature-based detection

+ Building of predictive model

Training set

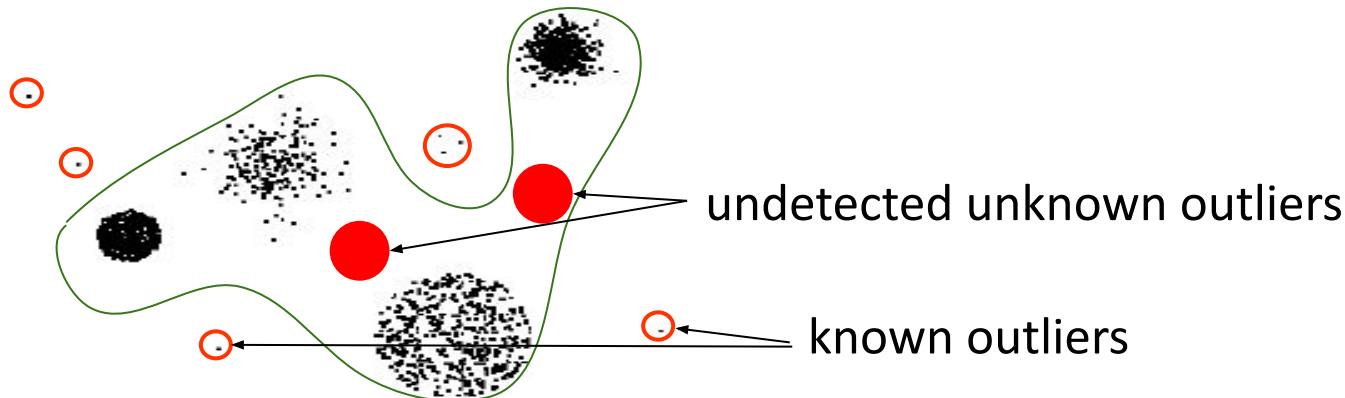
Tid	SrcIP	Start time	Dest IP	Dest Port	Number of bytes	Attack
1	206.135.38.95	11:07:20	160.94.179.223	139	192	No
2	206.163.37.95	11:13:56	160.94.179.219	139	195	No
3	206.163.37.95	11:14:29	160.94.179.217	139	180	No
4	206.163.37.95	11:14:30	160.94.179.255	139	199	No
5	206.163.37.95	11:14:32	160.94.179.254	139	19	Yes
6	206.163.37.95	11:14:35	160.94.179.253	139	177	No
7	206.163.37.95	11:14:36	160.94.179.252	139	172	No
8	206.163.37.95	11:14:38	160.94.179.251	139	285	Yes
9	206.163.37.95	11:14:41	160.94.179.250	139	195	No
10	206.163.37.95	11:14:44	160.94.179.249	139	163	Yes

Test set

Tid	SrcIP	Start time	Dest IP	Number of bytes	Attack
1	206.163.37.81	11:17:51	160.94.179.208	150	No
2	206.163.37.99	11:18:10	160.94.179.235	208	No
3	206.163.37.55	11:34:35	160.94.179.221	195	Yes
4	206.163.37.37	11:41:37	160.94.179.253	199	No
5	206.163.37.41	11:55:19	160.94.179.244	181	Yes

Training/building process

Predictive model
Decision tree
Neural network



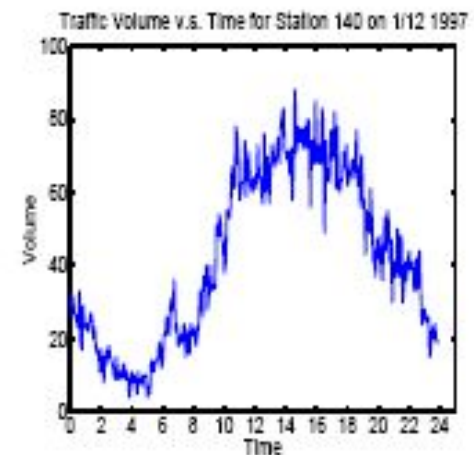
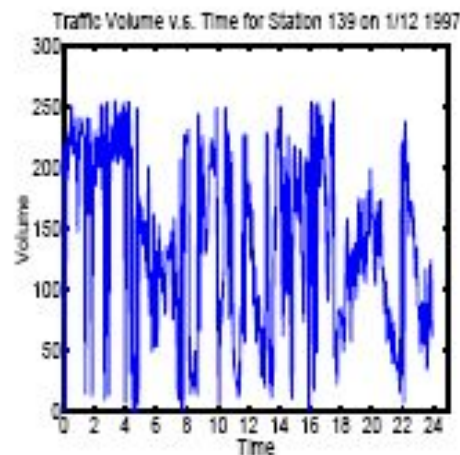
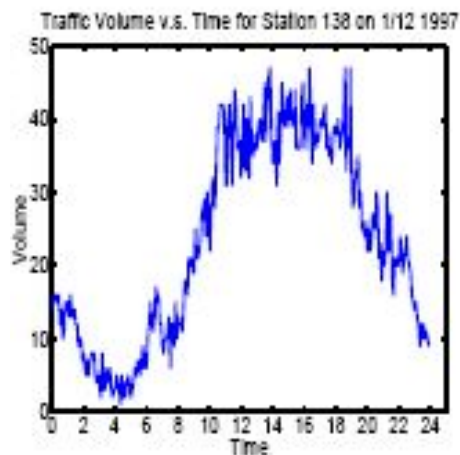
Anomaly Detection

- + Try to detect deviating data patterns by comparing the trends for data items that are close by (based on certain metric)
- + Example: Stations processing financial transactions
 - + Spatial attributes – time, location
 - + Non spatial attribute – volume (number of transactions)

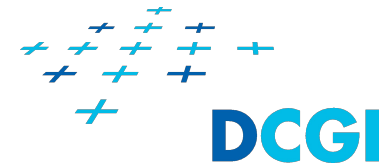
Three stations at close by locations

Change in volume over time is visualized

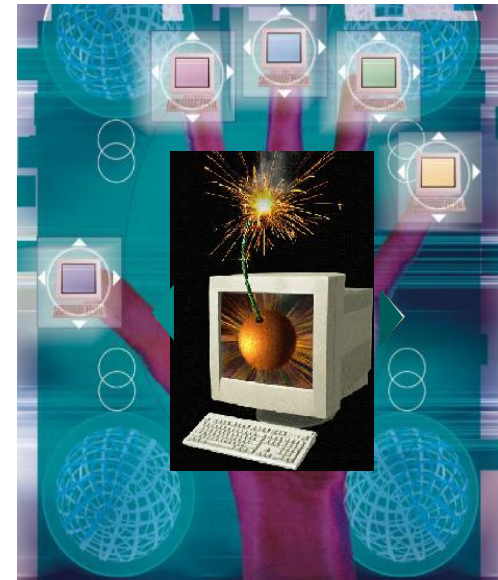
The trend for station in the middle is deviation (it is suspicious)



Signature-based Detection vs Anomaly Detection



- + **Intrusion Detection**
 - + Process of monitoring the events occurring in a computer system or network and analyzing them for intrusions
 - + Intrusions are defined as attempts to bypass the security mechanisms of a computer or network
- + **Challenges**
 - + Traditional signature-based detection systems are based on signatures of known attacks and may not detect emerging cyber threats
 - + Substantial latency in deployment of newly created signatures across the computer system
- + Anomaly detection can alleviate these limitations



Applications of Anomaly Detection

- + **Fraud detection** refers to detection of criminal activities occurring in commercial organizations
 - + Malicious users might be the actual customers of the organization or might be posing as a customer (also known as identity theft).
- + **Types of fraud**
 - + Credit card fraud
 - + Insurance claim fraud
 - + Mobile / cell phone fraud
 - + Insider trading
- + **Challenges**
 - + Fast and accurate real-time detection
 - + Misclassification cost is very high

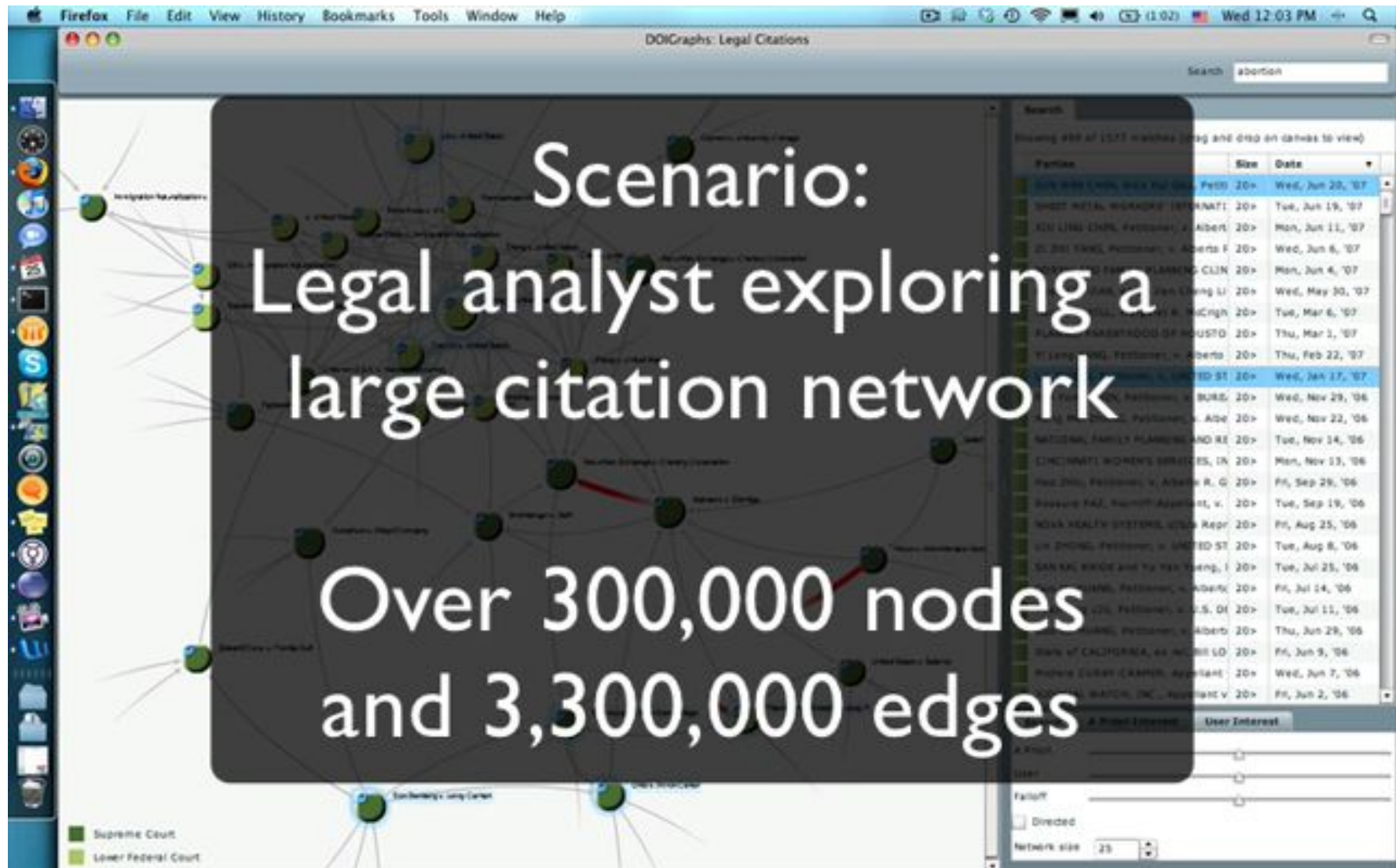


- + Overview first, filter and zoom, then details on demand
- + In outlier/intrusion detection the visualization mantra does not make sense
 - + There is no need to visualize overview of all data if a data mining technique detects suspicious data item
 - + Instead, an analyst will typically visualize the item and explore its context to confirm or dismiss the suspicion
- + Alternative visualization mantra
 - + Search, Show Context, Expand on Demand [van Ham and Perer 09]

EXAMPLE OF VISUAL ANALYTICS

Search, Show Context, Expand on Demand

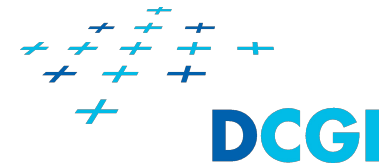
[van Ham and Perer 09]



SocialAction:
Analyzing the
Social Network
of US Senators

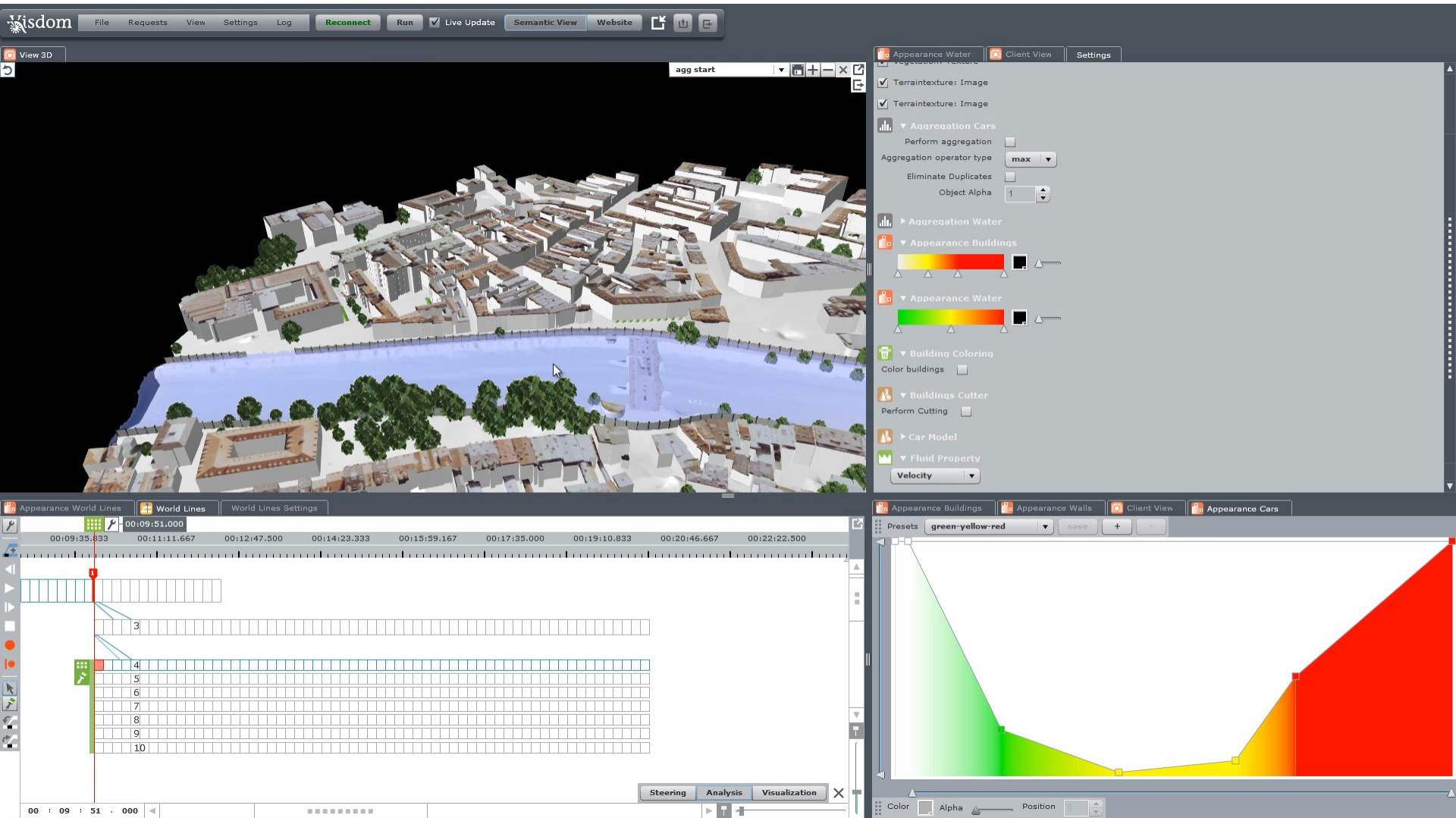
Adam Perer
Ben Shneiderman

Visual analysis and steering of ensemble simulations

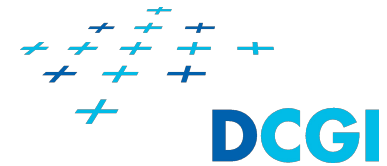


- + Scenario
 - + Flooding of Vienna
- + Multiple different variants
 - + The barrier can breach on multiple different locations
- + Question
 - + What is the probability of damage made when the barrier breaches
- + The analyst can be
 - + Town hall member - s/he is deciding if and where the barrier should be made stronger
 - + Analyst of insurance company - s/he is deciding what should the price of the insurance against flooding be for certain parts of the city

Visual analysis and steering of ensemble simulations

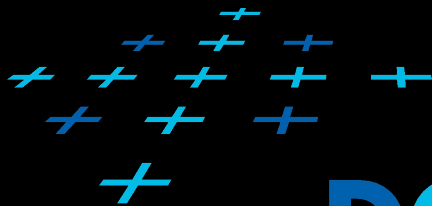


Summary



- + Introduction
 - + What is visual analytics
- + Big data
 - + What are big data
 - + V3(4)
 - + Volume, Variety, Velocity, (Veracity)
 - + Challenges in big data visualization
 - + Some solutions
- + Automated analysis techniques + visualization
- + Visual analytics examples

Thank you for your attention



DCGI

